

A Resilient and Power-Efficient Automatic-Power-Down Sense Amplifier for SRAM Design

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EC 571: Digital VLSI Circuit Design — Final Project

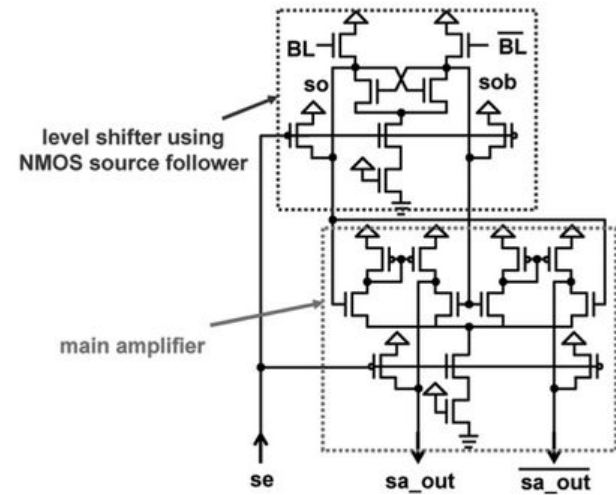
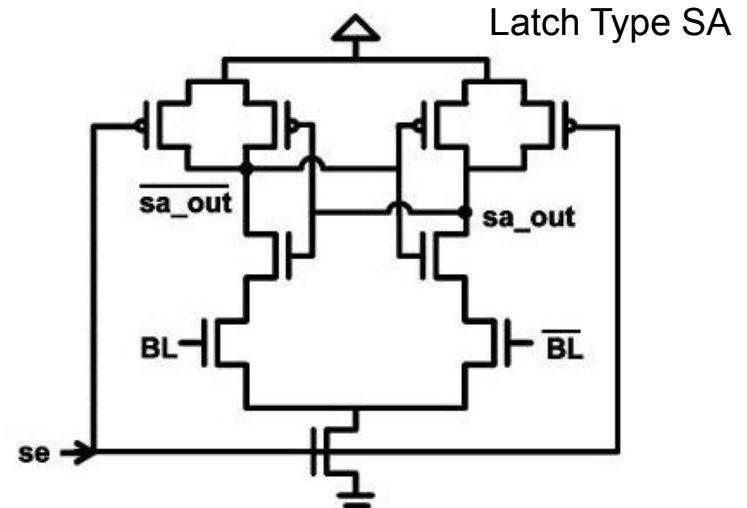
Based on: Y.-C. Lai and S.-Y. Huang, "A Resilient and Power-Efficient Automatic-Power-Down Sense Amplifier for SRAM Design," IEEE TCAS-II, vol. 55, no. 10, pp. 1031–1035, Oct. 2008.

Outline

- 6T SRAM cell basics
 - Latch-type sense amplifier
 - Current mirror SA
 - Current Mirror SA with power down
 - Power Comparison
-
- APD block
 - Schmitt trigger
-
- Dual-V_{HL} APD circuitry

Motivation

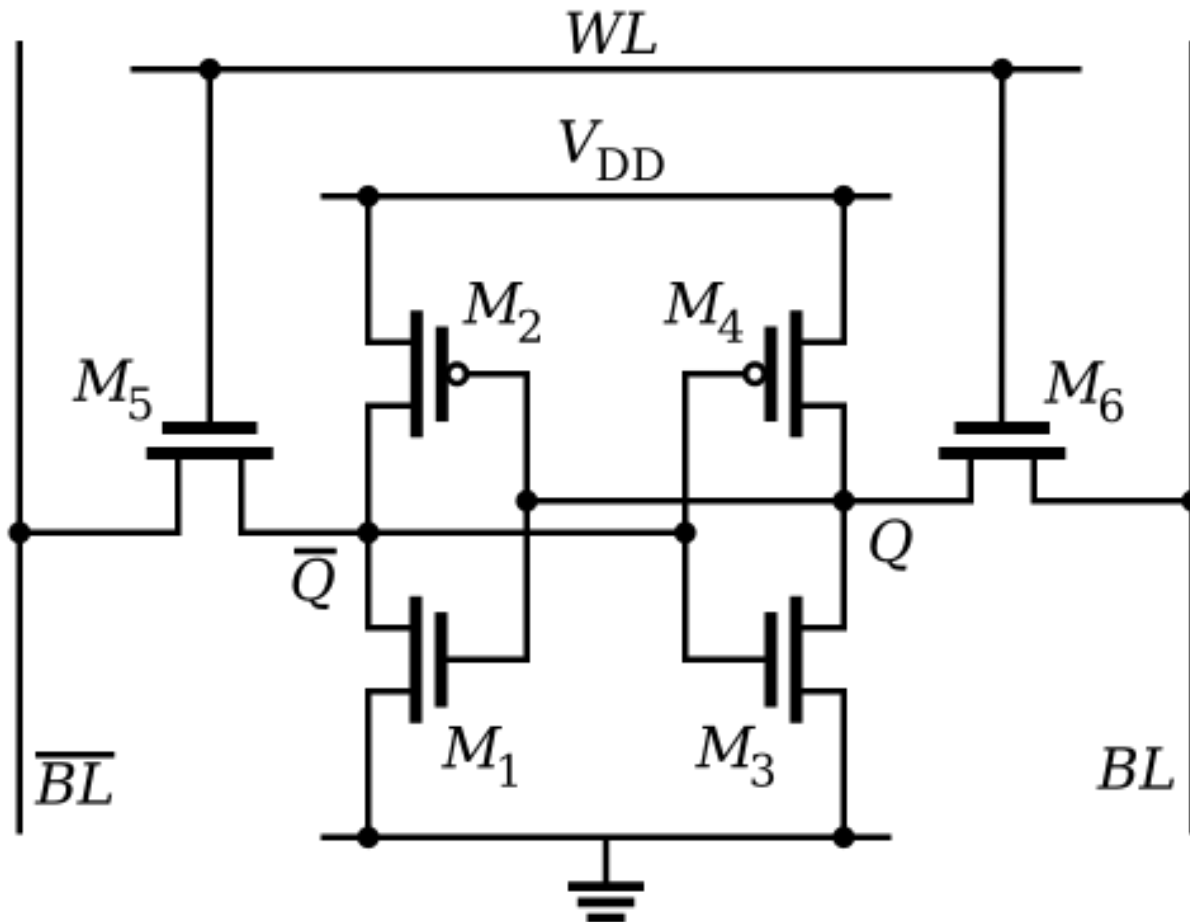
- SRAM sense amplifiers detect small BL / BLBAR swings
- Latch-type SA: fast & low-power, but **vulnerable to sensing failure under V_t variation**
 - “malfunction of the sense amplifier due to insufficient sensing margin against the input offset voltage.”
- Current-mirror SA: consumes **too much power**
- **Paper's idea:** create a current mirror SA which consumes less power (powers down)



Current Mirror SA

6T SRAM Cell with Precharge

Write 1, read 1, write 0, read 0

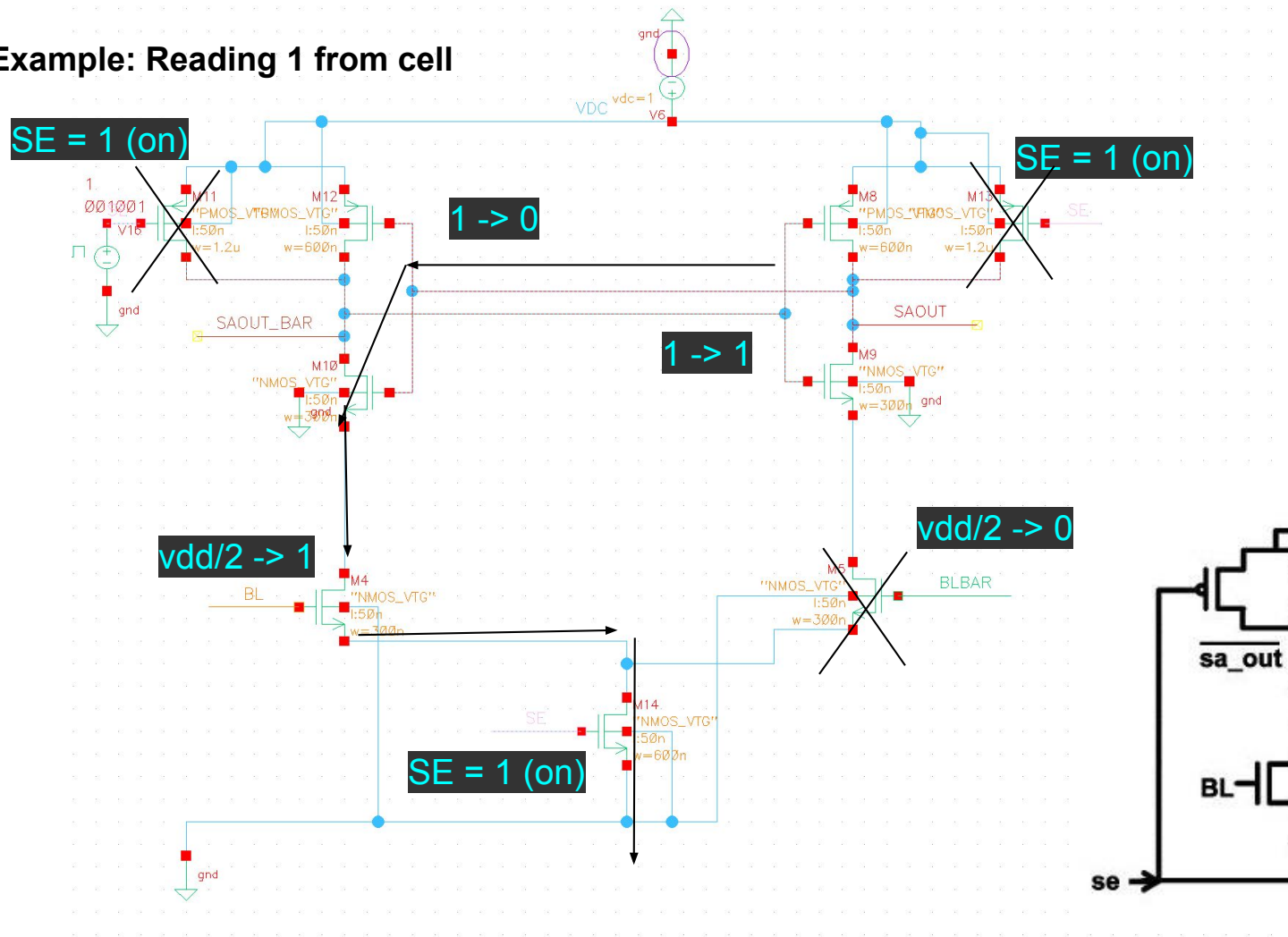


Design parameters

- $V_{DD} = 1\text{ V}$
- Unit NMOS: $L = 50\text{ n}$, $W = 90\text{ n}$
- Unit PMOS: $L = 50\text{ n}$, $W = 180\text{ n}$
- Access NMOS: $W = 120\text{ n}$
- Write & Precharge PMOS: $W = 10\text{ u}$
- Write Discharge NMOS: $W = 5\text{ u}$

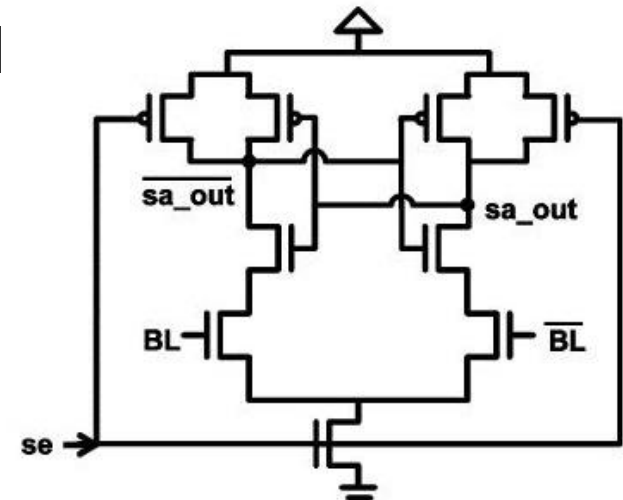
Latch Type Sense Amplifier for 6T SRAM

Example: Reading 1 from cell

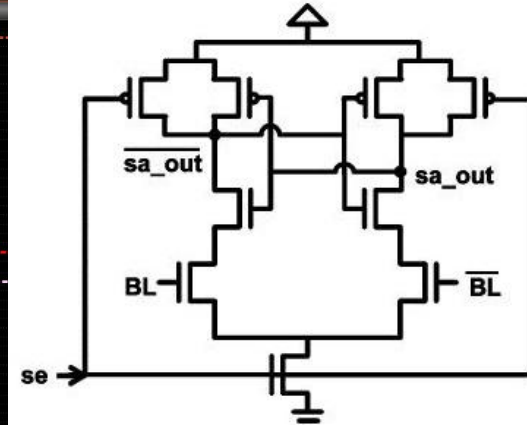
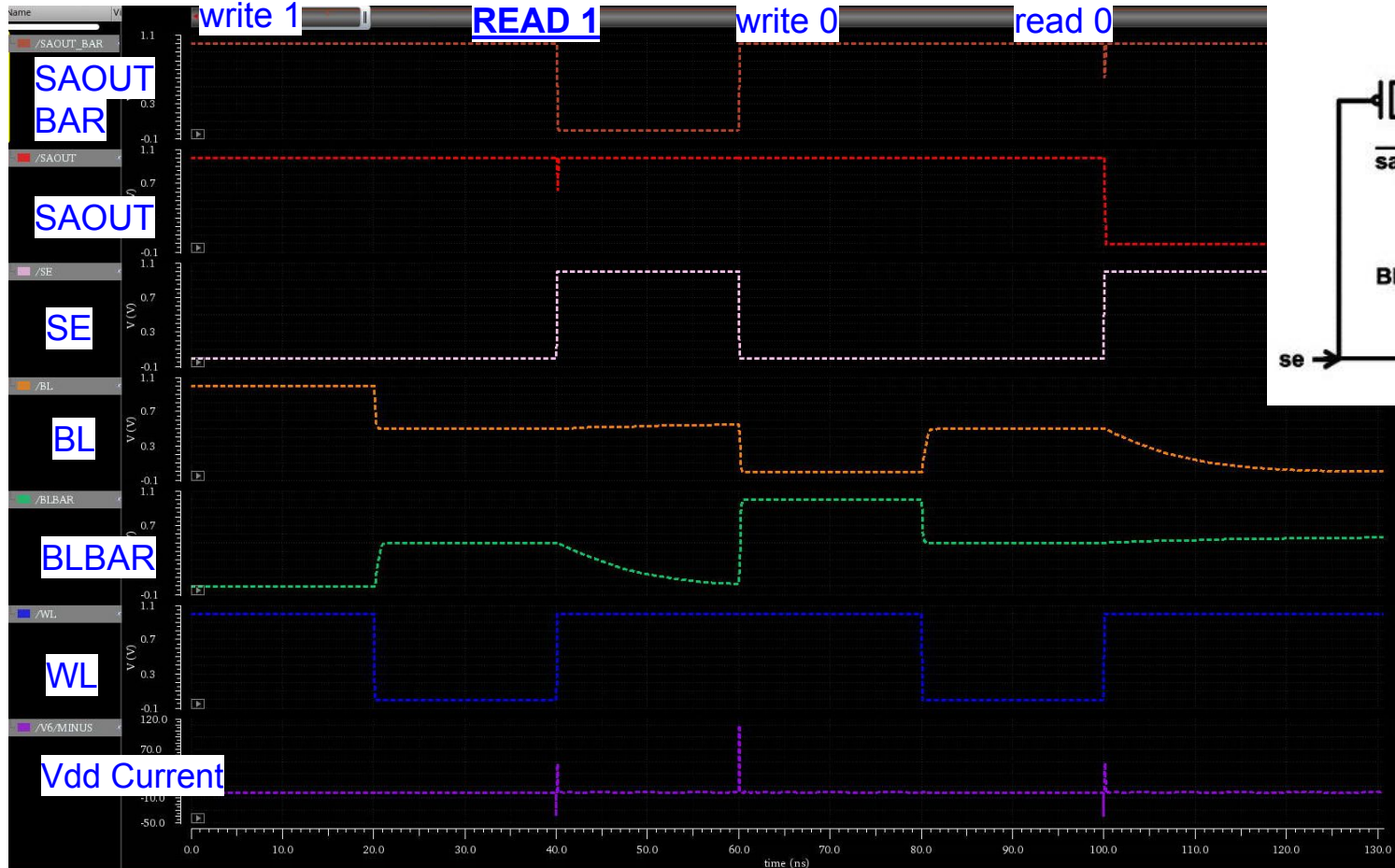


Sizes:

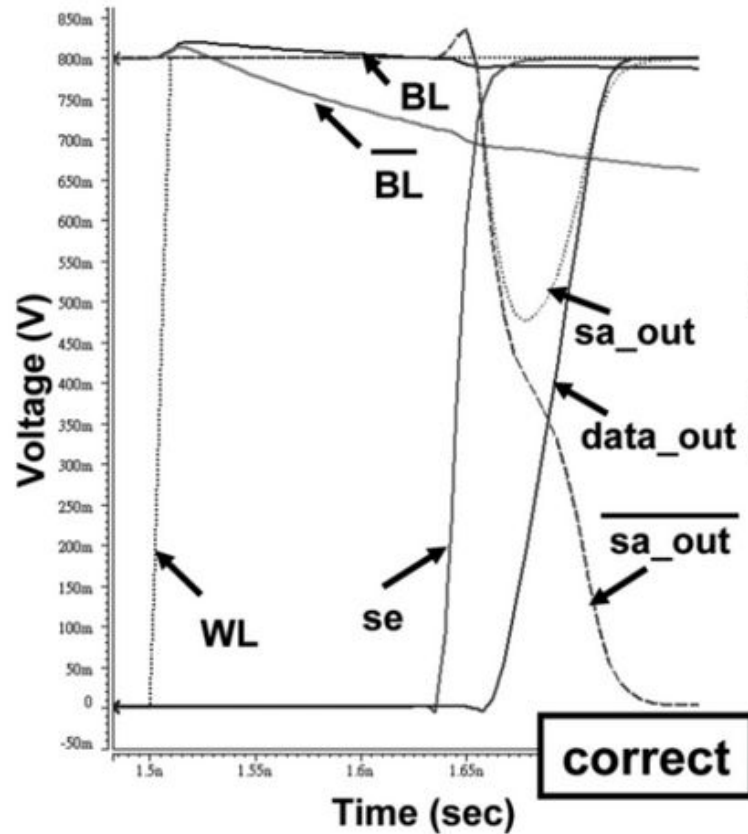
Total "Silicon Area Cost" =
40 Unit Transistors



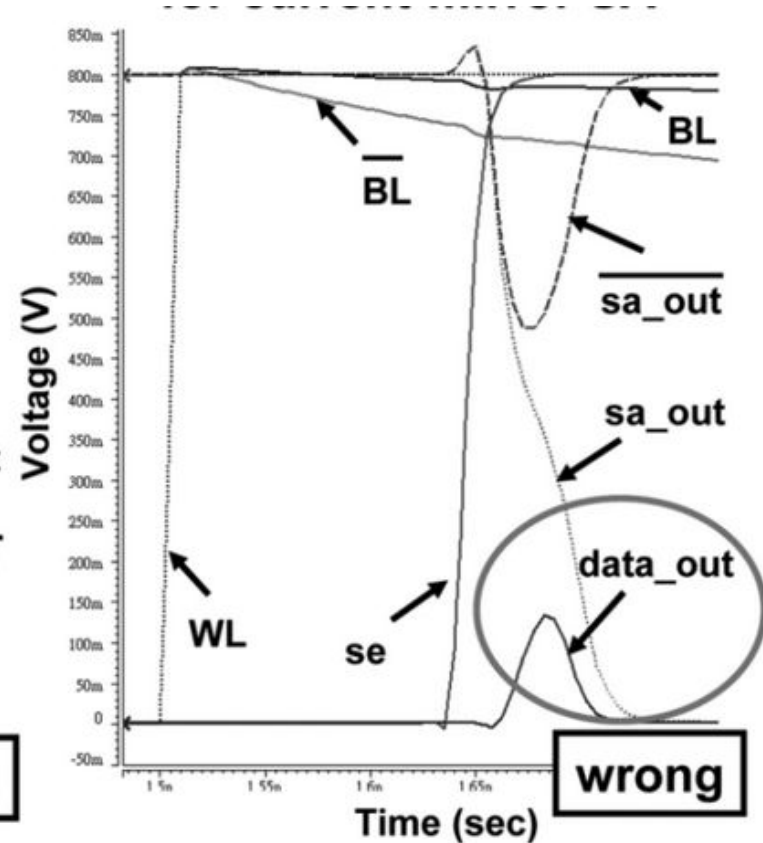
6T + Latch-Type SA waveform



Vt Variation Failure

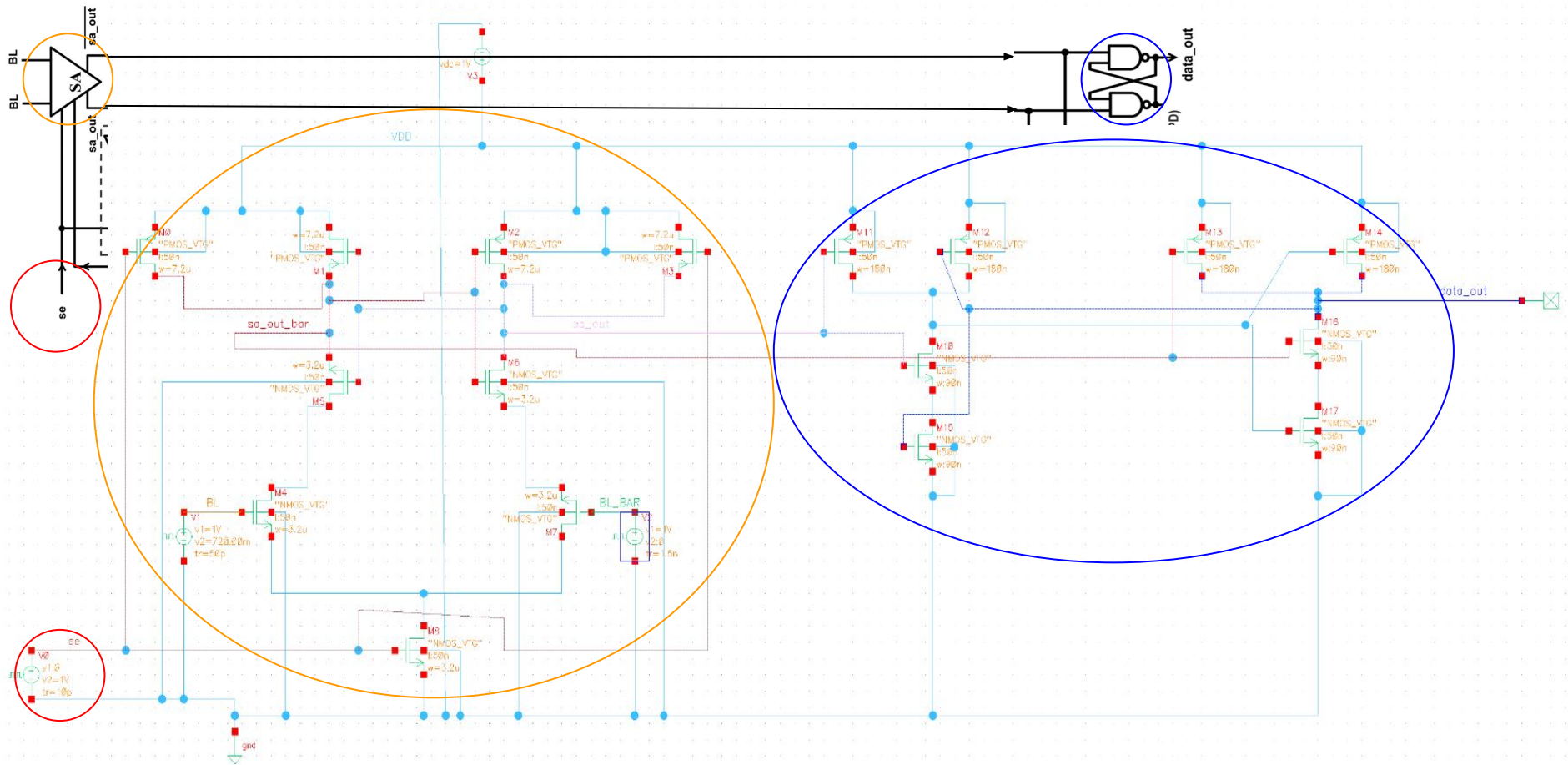


(c) large bitline swing
for latch-type SA



(d) small bitline swing
for latch-type SA

Vt Variation Failure - test

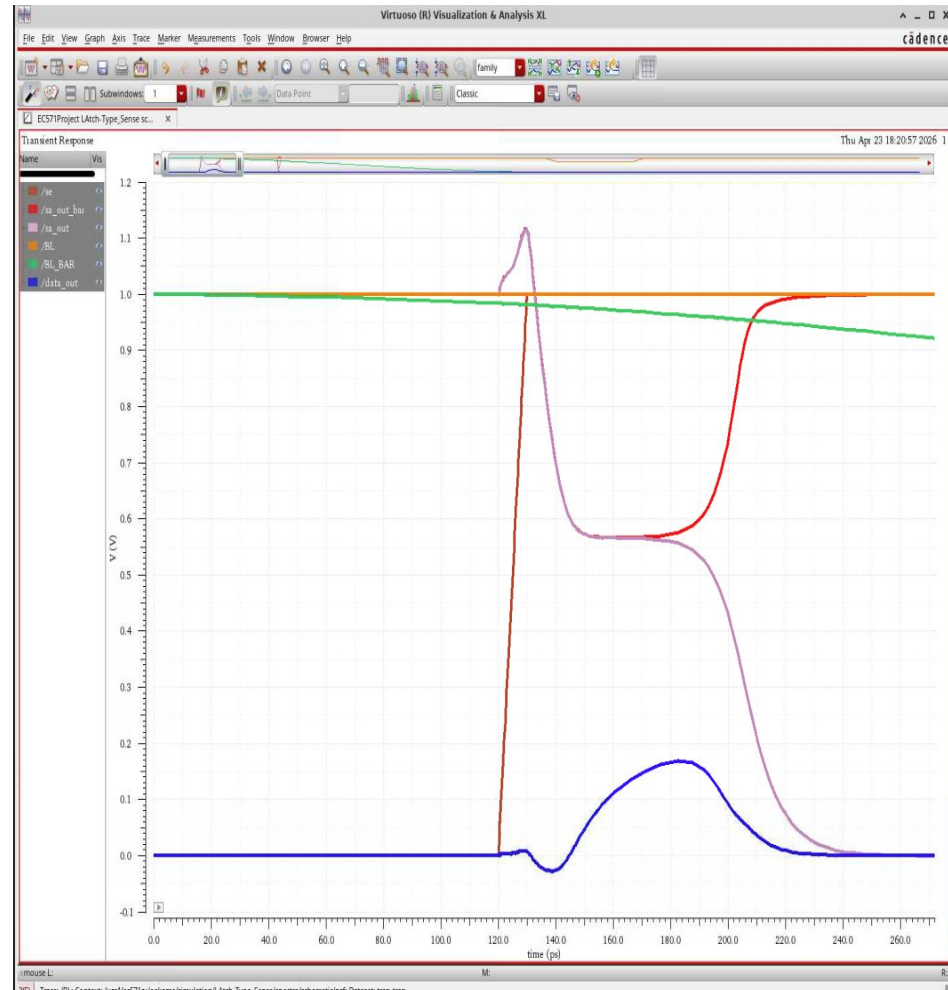
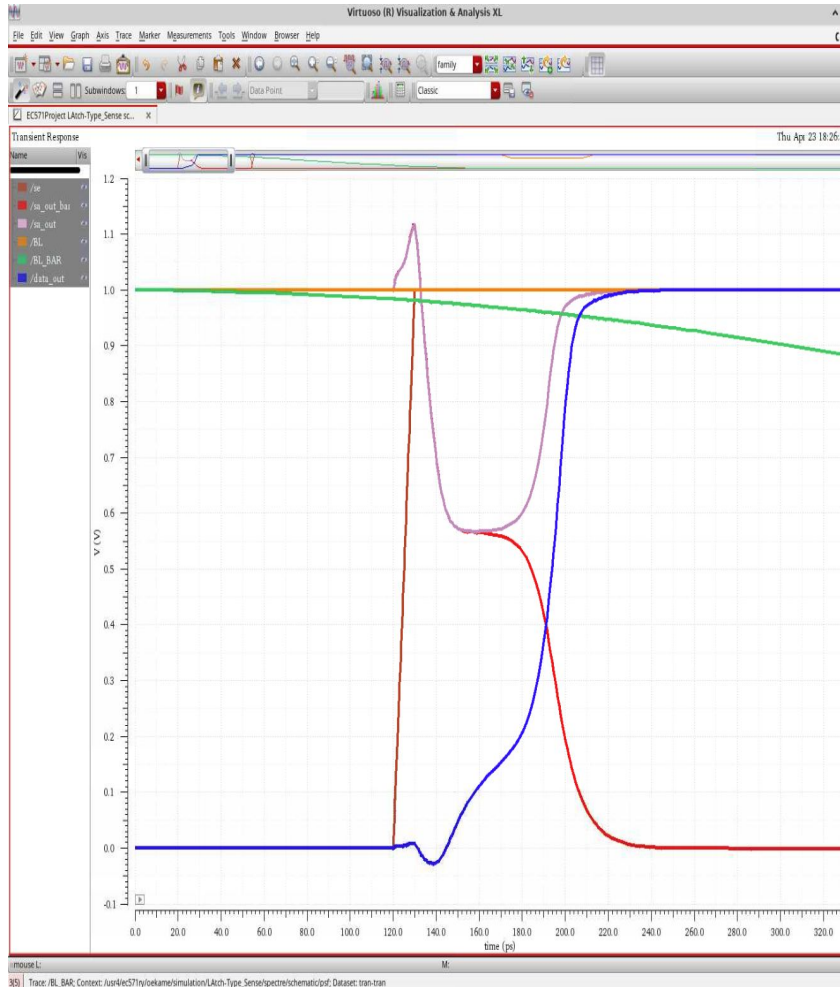


Vt Variation Failure - test

$$\Delta V_{BL} = \frac{t_{SE}}{TR} \times (V_1 - V_2)$$

$$\Delta V_{BL} = \frac{100 \text{ ps}}{1510 \text{ ps}} \times (1 - 0.72)$$

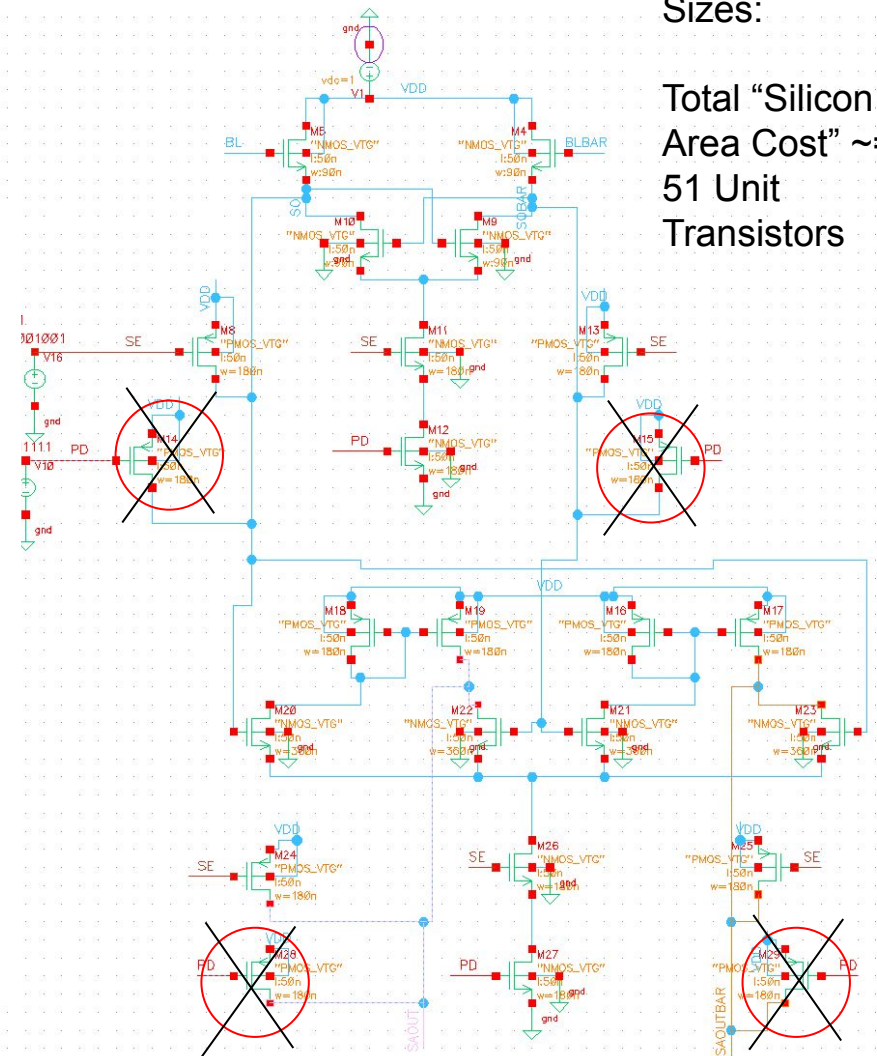
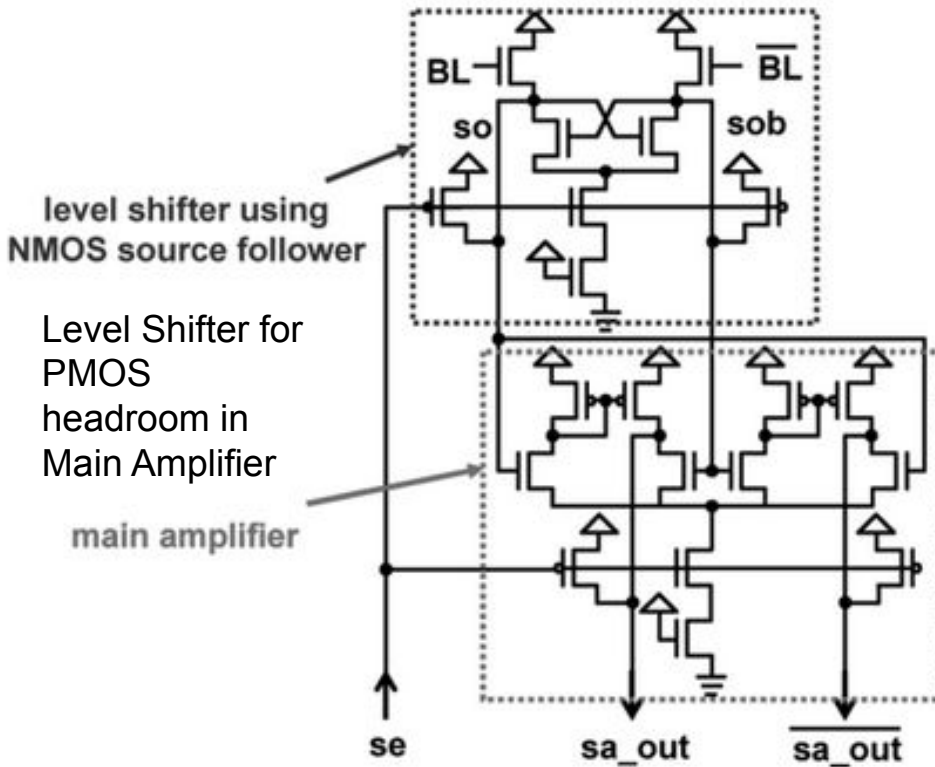
$$\Delta V_{BL} = 0.066 \times 0.28 \approx \mathbf{18 \text{ mV}}$$



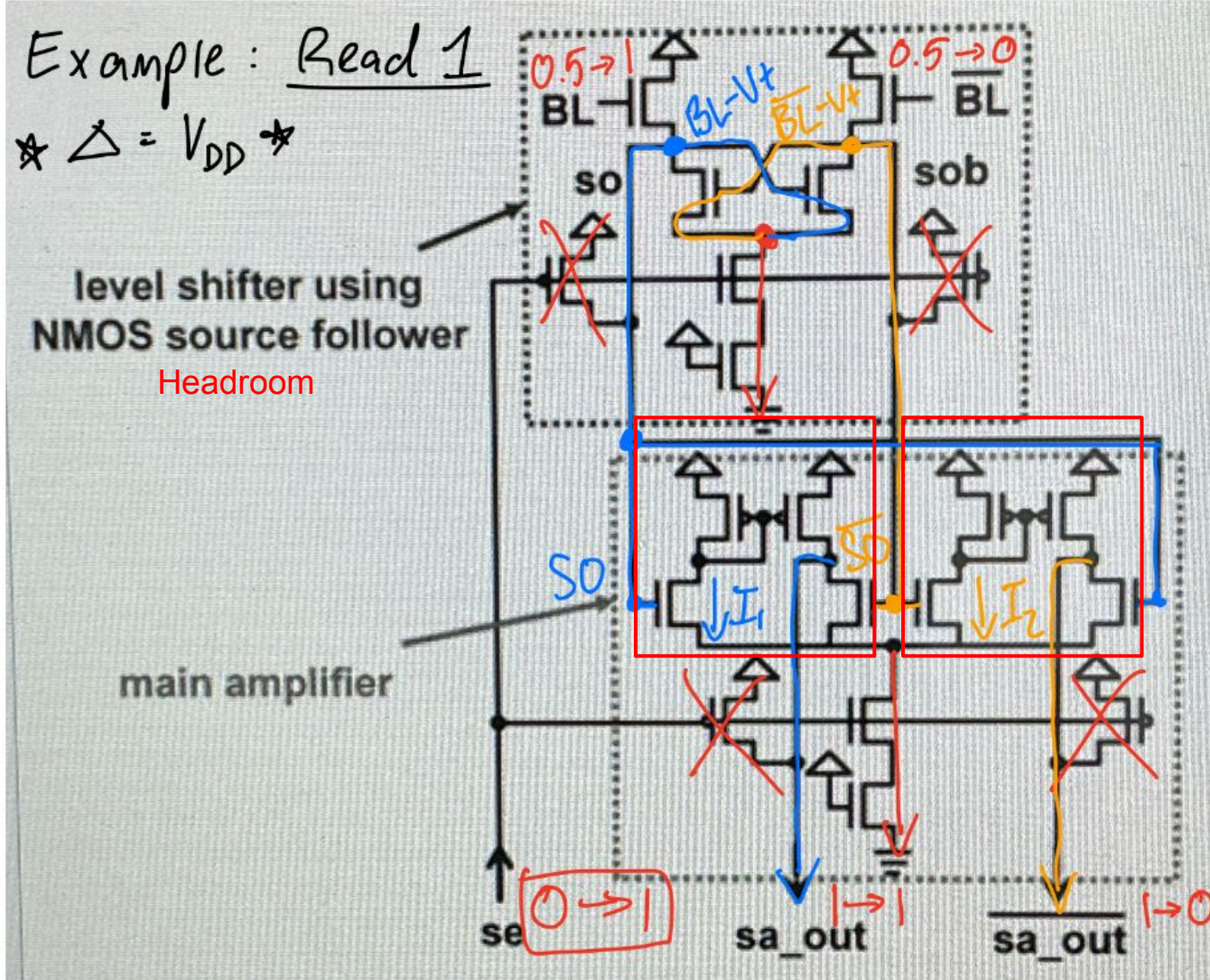
Current Mirror SA design for 6T (without Power Down)

Sizes:

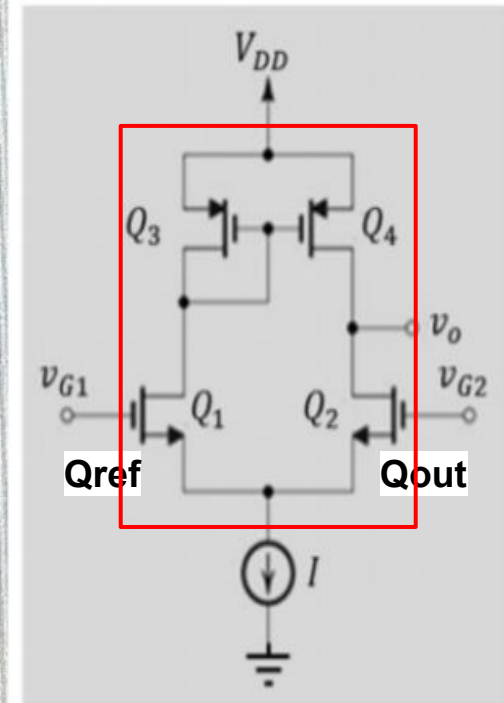
Total "Silicon Area Cost" $\sim$ 51 Unit Transistors



Current Mirror SA design for 6T (without Power Down)

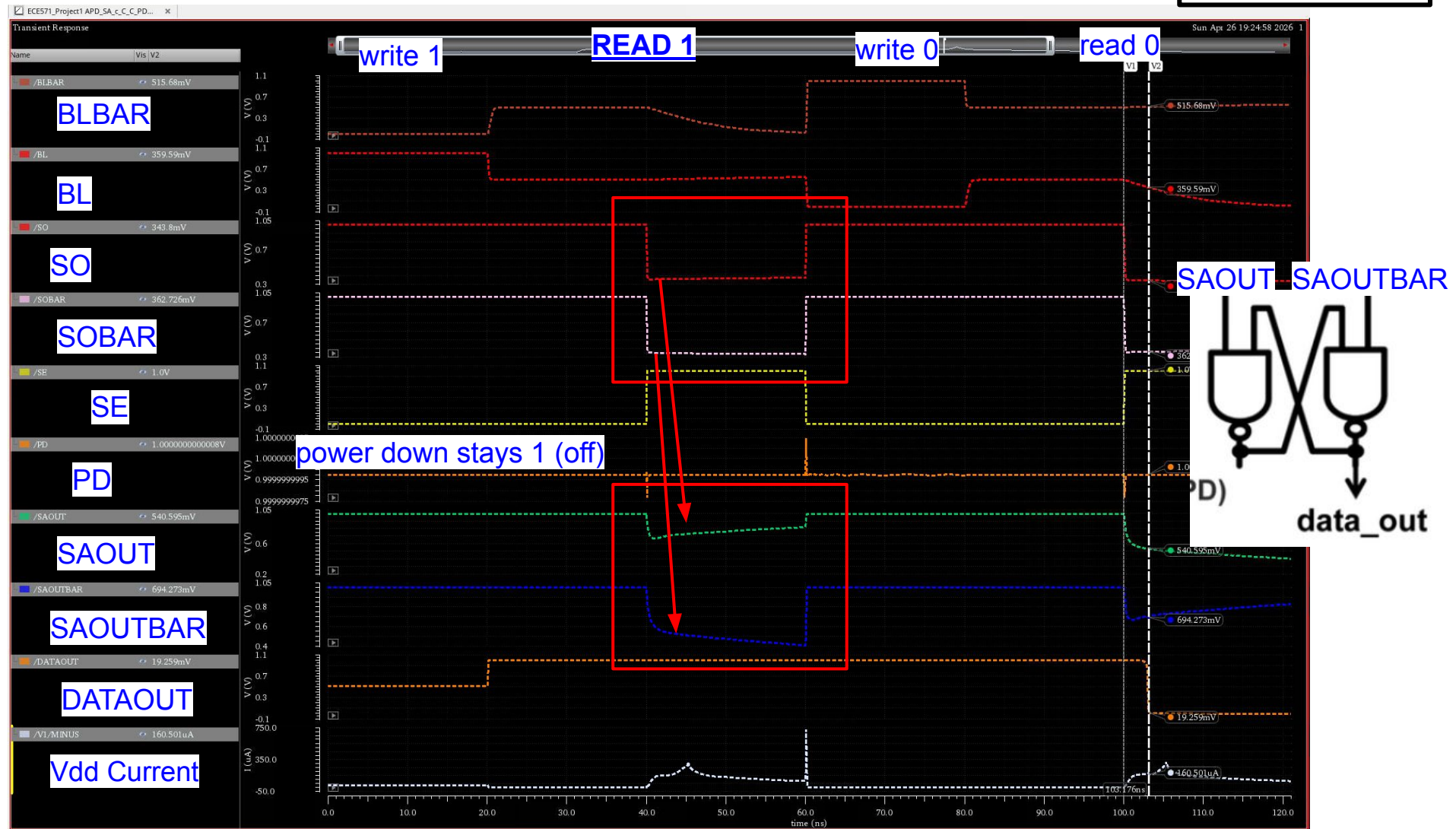


Current Mirror
Differential Amplifier



Current Mirror Output without Power Down

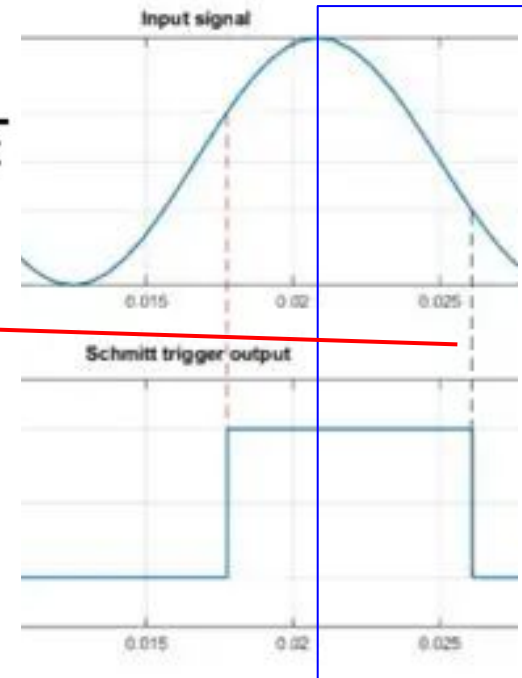
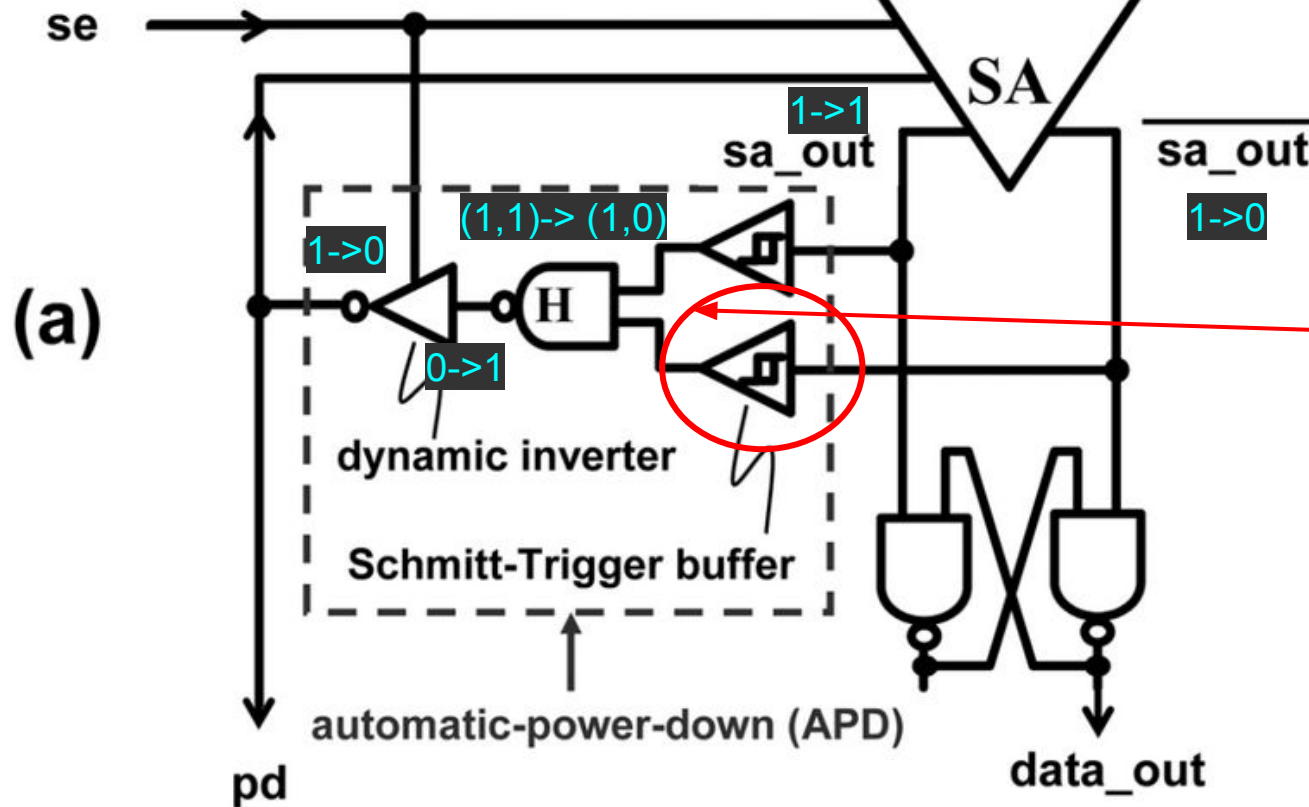
Power: 123.6 μ W
Delay: 3.1 ns



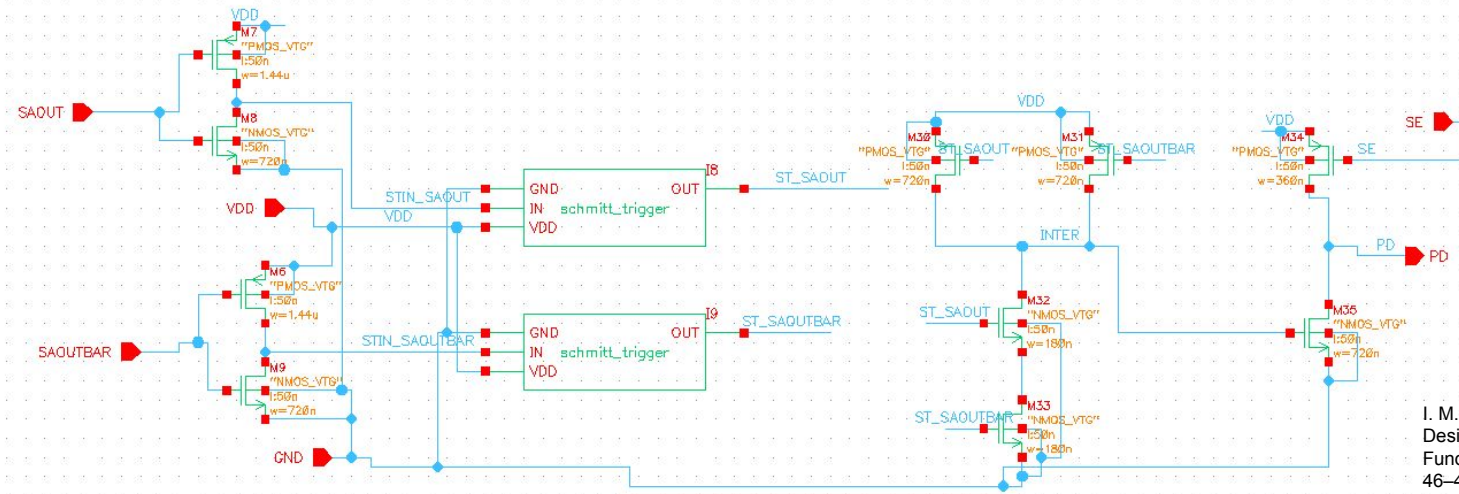
APD Block - Automatic Power Down

“It is notable that is easy to adjust VHL at the design stage by tuning the transistor sizes”

Example: READ 1



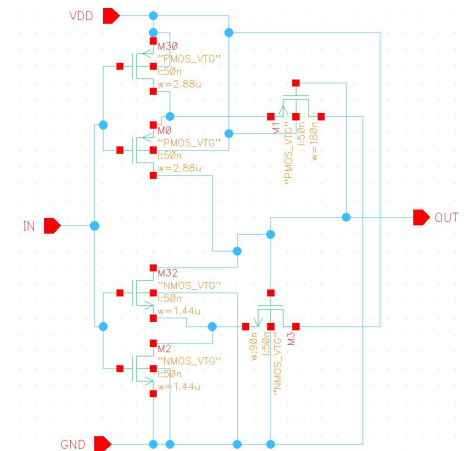
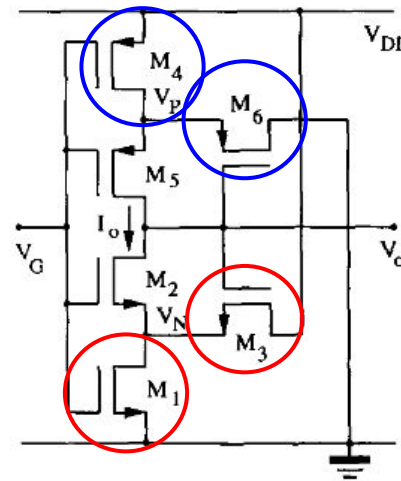
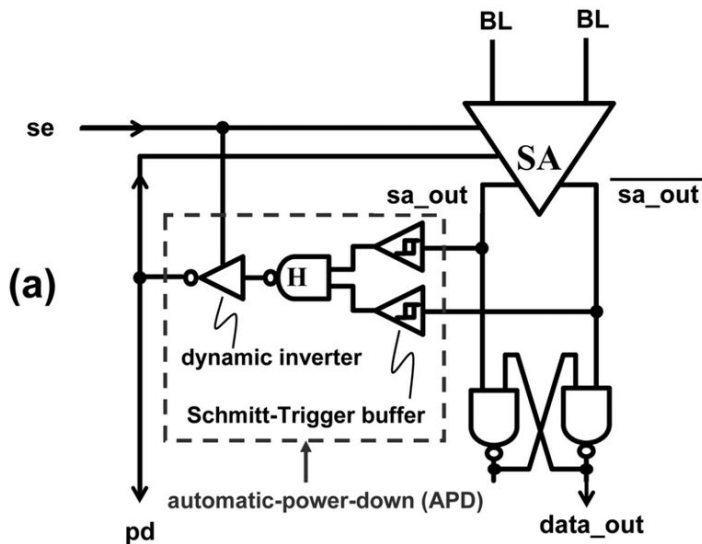
APD Block - Automatic Power Down - Our design



$$\frac{k_1}{k_3} = \left(\frac{V_{DD} - V_{Hi}}{V_{Hi} - V_{TN}} \right)^2$$

$$k_1/k_3 = 16 = k_4/k_6$$

I. M. Filanovsky and H. Bakes, "CMOS Schmitt Trigger Design," IEEE Transactions on Circuits and Systems I: Fundamental Theory and Applications, vol. 41, no. 1, pp. 46-49, Jan. 1994.



Current Mirror SA design for 6T (with Power Down)

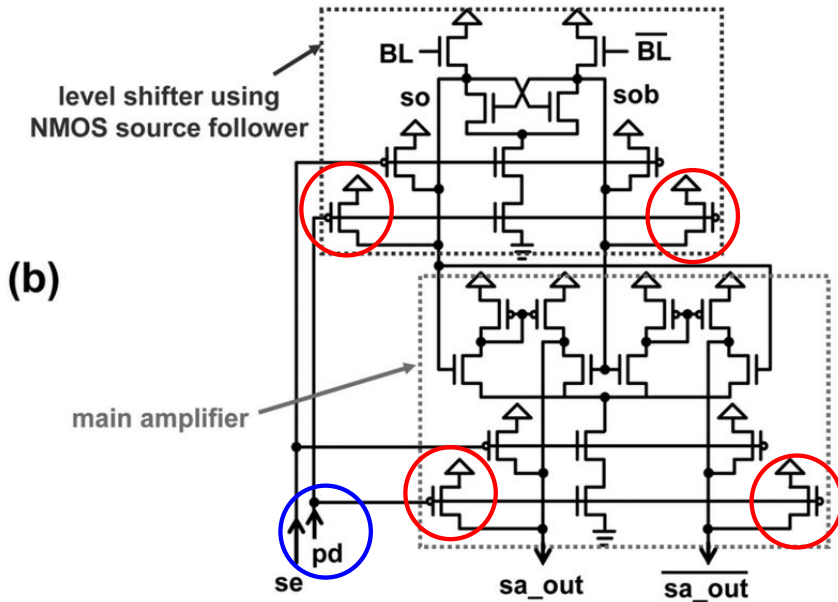
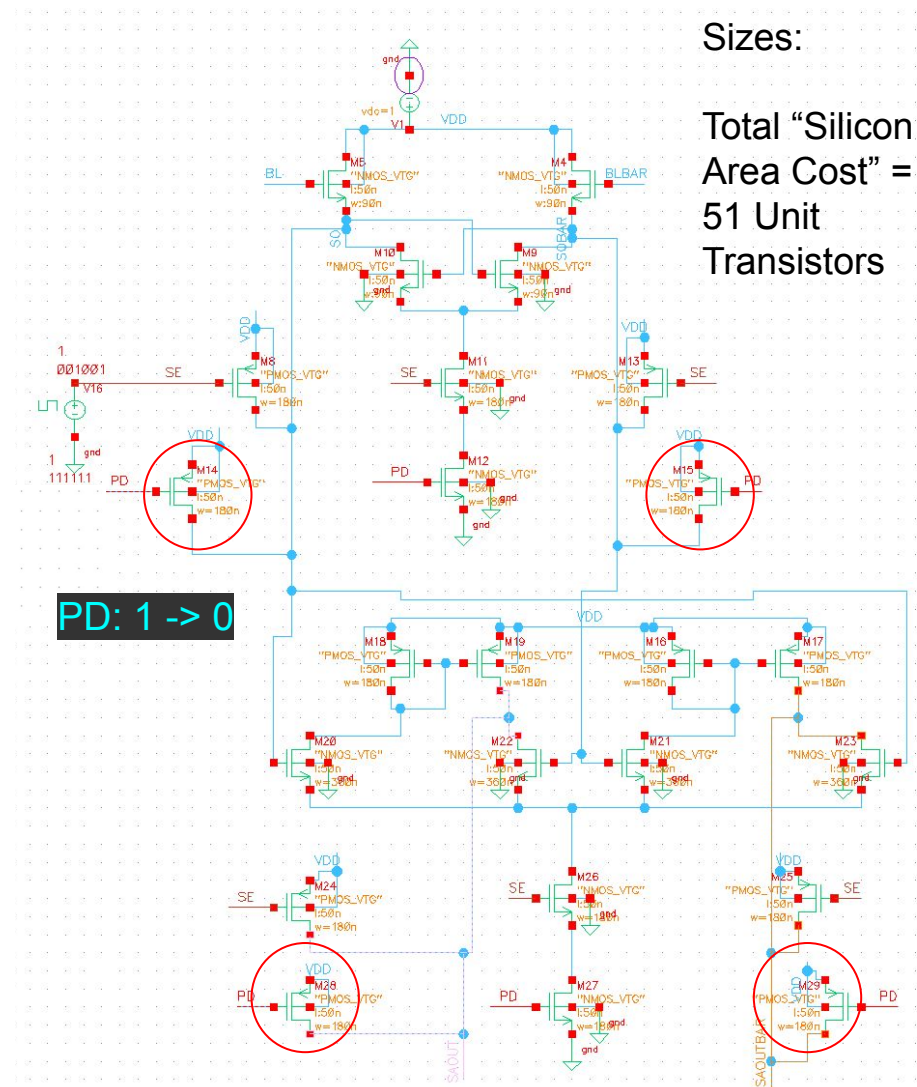
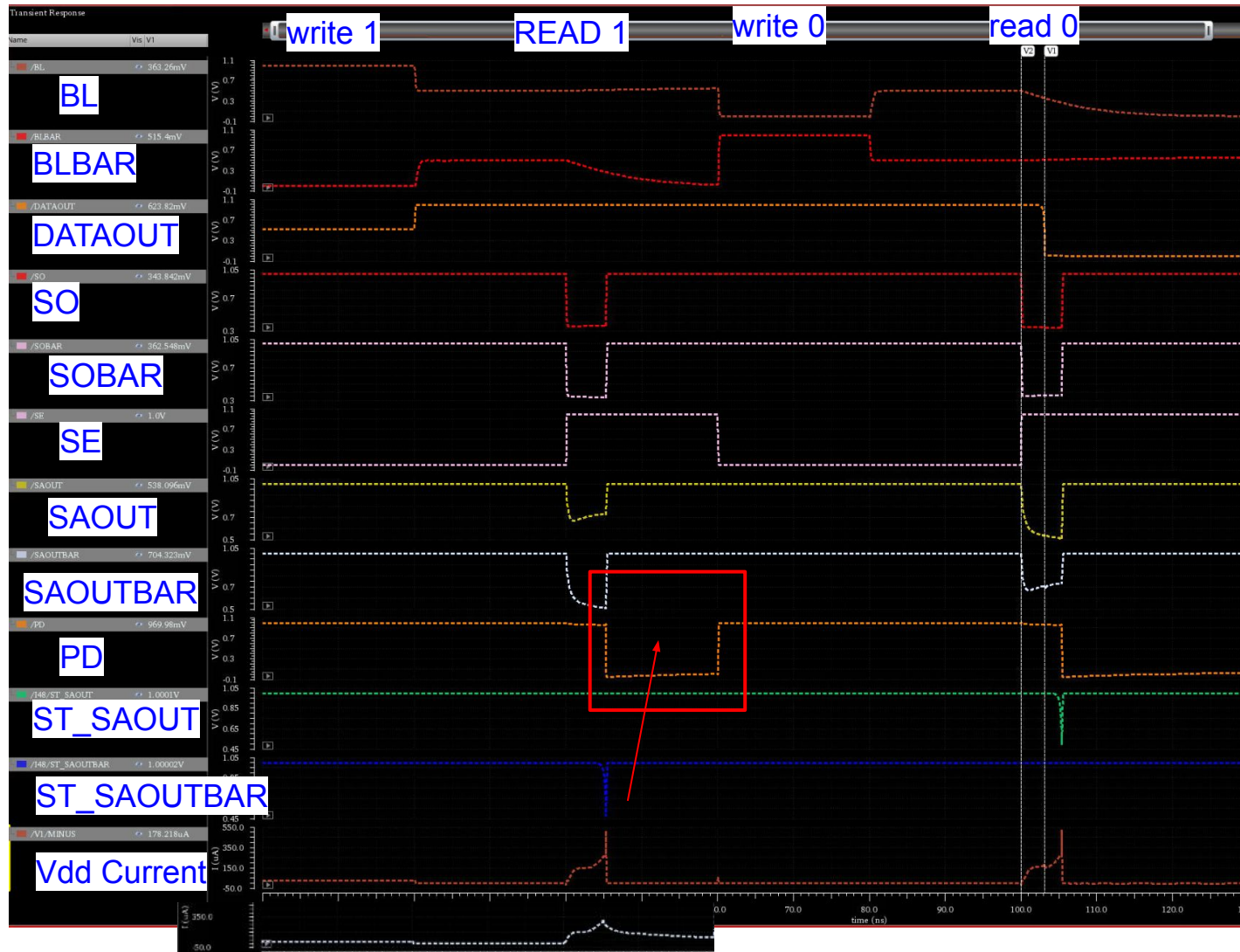


Fig. 5. Proposed APD sense amplifier. (a) Overall architecture. (b) Detailed circuit diagram of the sense amplifier.

Pulls up SO/SOBAR, SAOUT/SAOUTBAR



Current Mirror Output with Power Down



Power:
22.9 μ W
Delay:
3.1 ns

Question:
Why Power
Down?

Current
Comparison:
PD vs No PD
(overlay)

With/Without APD - Power & Delay Comparison

Design	Power (uW)	Delay (ns)
Power Down	23	3.1
No PD	124	3.1

Relationships ~**same as paper**

- different exact values due to technology / sizings / design choices

Dual- V_{HL} APD

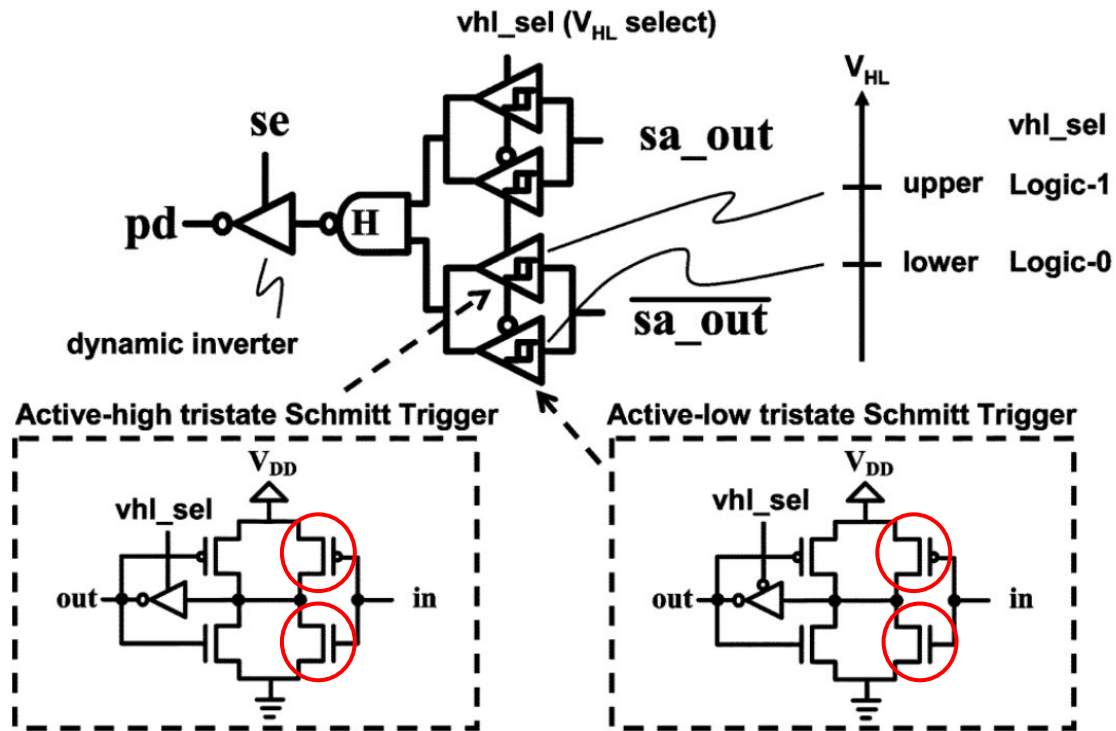


Fig. 9. Architecture of dual- V_{HL} APD circuitry.

Dual-V_{HL} APD

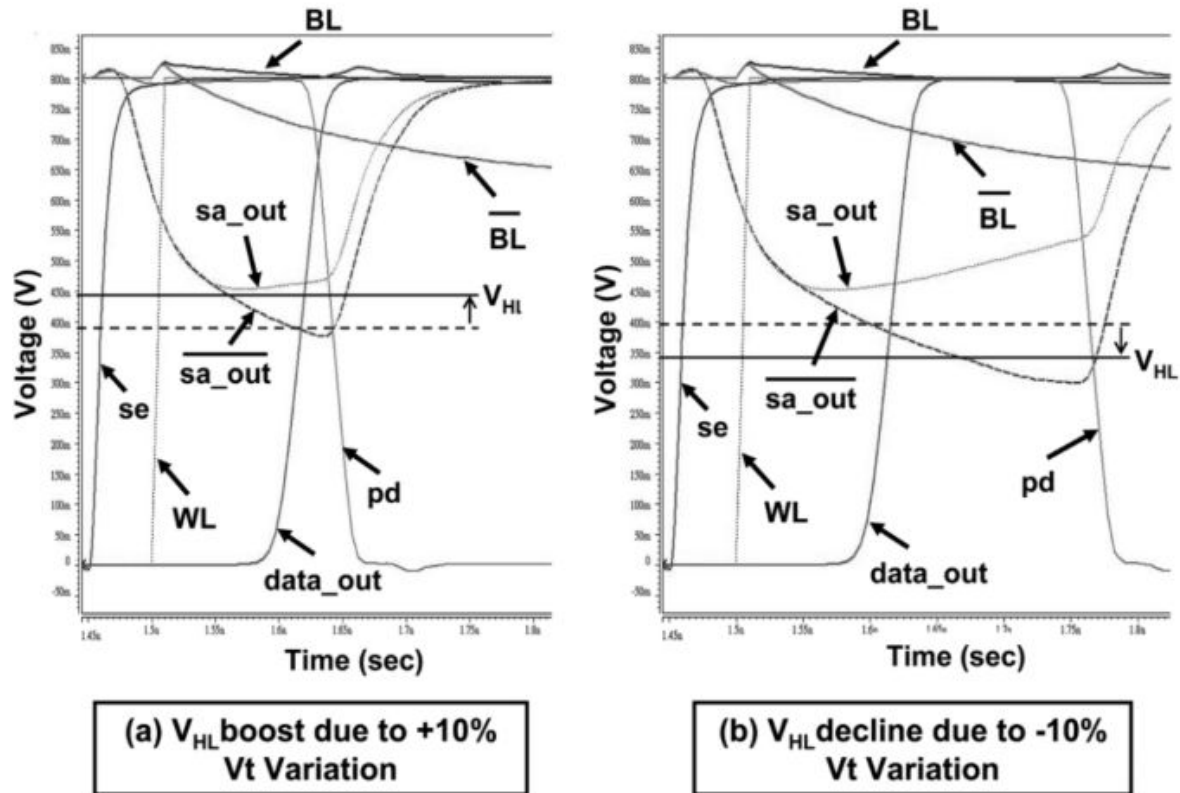
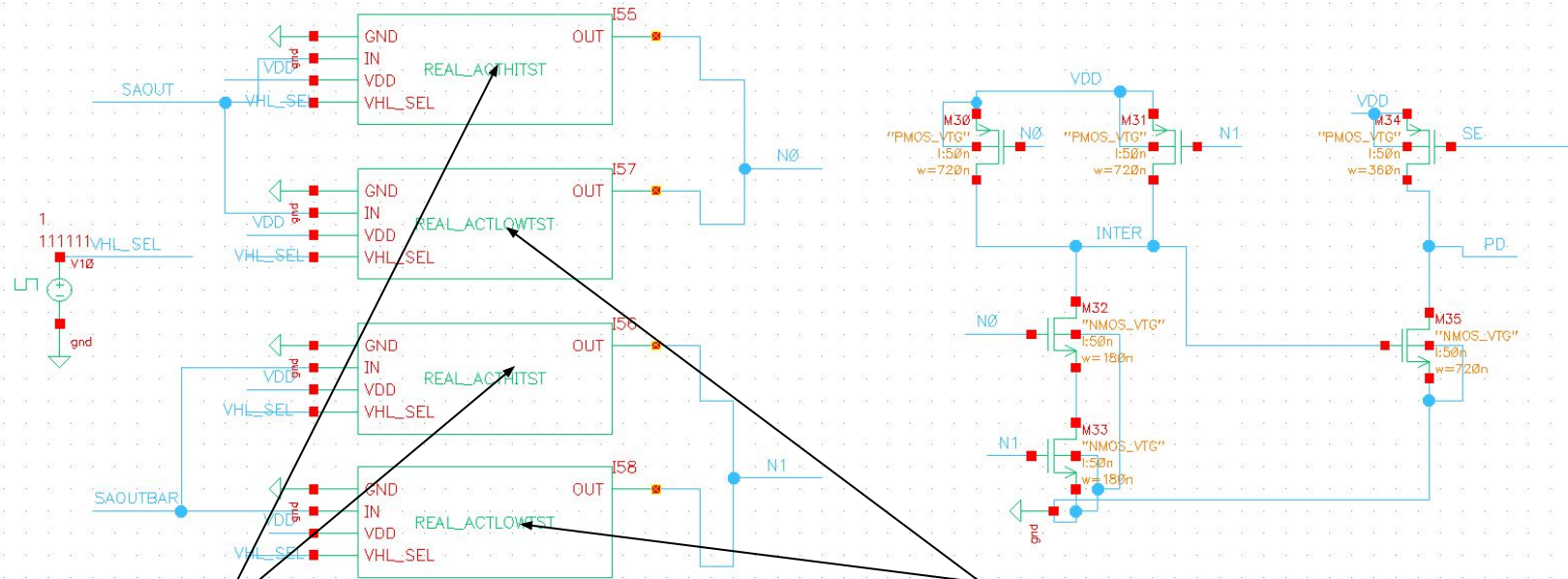
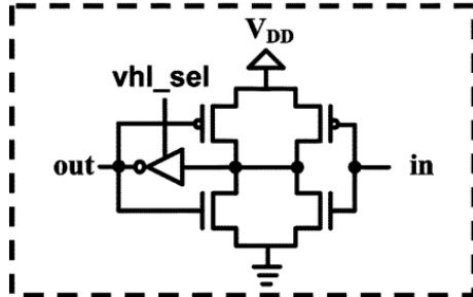


Fig. 7. Simulation waveforms of proposed sense amplifier under (a) +10% and (b) -10% V_t variation imposed on APD.

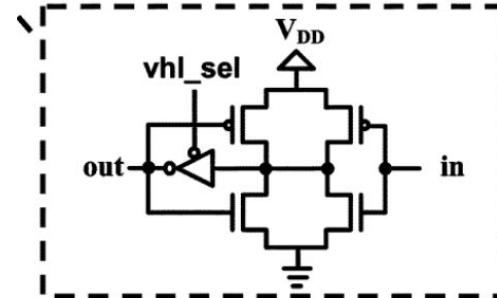
Dual-V_HL APD



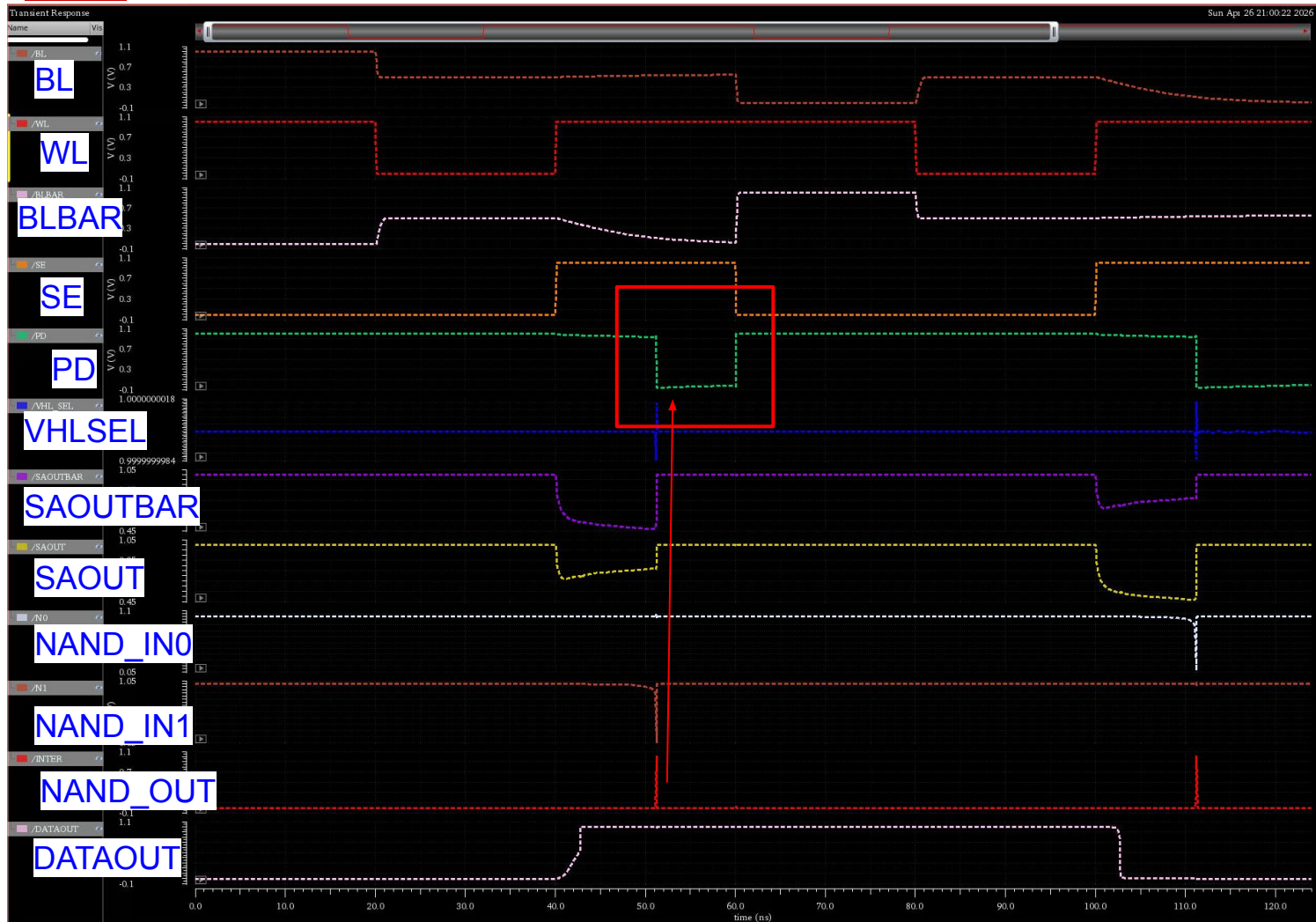
Active-high tristate Schmitt Trigger



Active-low tristate Schmitt Trigger



Dual-V_HL APD



References

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Thank You / Questions?